INCH-POUND

MIL-C-39006/28

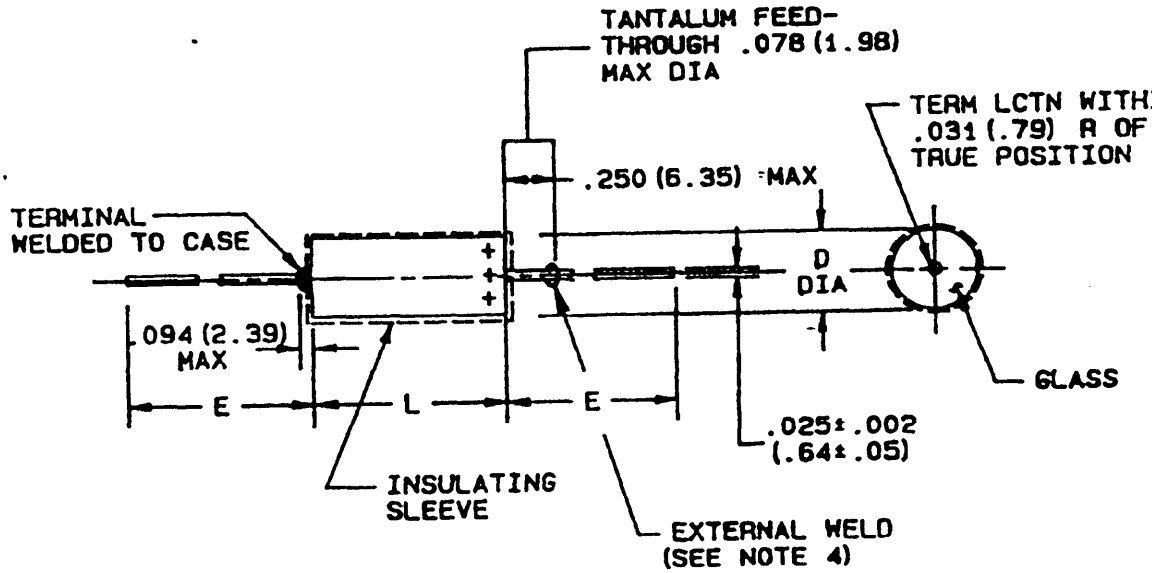
25 September 1992

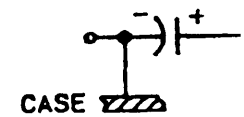
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), HIGH RELIABILITY, STYLE CLS79

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39006.





CIRCUIT DIAGRA

Inch	mm
.002	0.05
.025	0.64
.031	0.79
.078	1.98
.094	2.39
.250	6.35

Case size	Dimensions			
	Basic case		Insulated case	
	L 1/ +.031 (0.79) -.016 (0.41)	D ±.016 (0.41)	D max	E ±.250 (6.35)
T1	.453 (11.51)	.188 (4.78)	.219 (5.56)	1.500 (38.10)
T2	.641 (16.28)	.281 (7.14)	.312 (7.92)	2.250 (57.15)
T3	.766 (19.46)	.375 (9.52)	.406 (10.31)	2.250 (57.15)
T4	1.062 (26.97)	.375 (9.52)	.406 (10.31)	2.250 (57.15)

1/ Length of basic case sleeving shall be as specified in MIL-C-39006.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLS79 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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QUALIFICATION:

A manufacturer shall be considered for qualification to style CLS79 if currently qualified to the R or S failure rate level, as applicable, of MIL-C-39006 for the equivalent "H" designated units of style CLR79 capacitors.

REQUIREMENTS:

Capacitors supplied to this specification sheet shall meet the requirements specified herein and shall be capable of meeting the requirements of the basic specification as demonstrated by the manufacturer's current qualification thereto.

Category A distributors: Category A distributors may be used under the following conditions:

a. The distributor shall be an authorized distributor of the manufacturer in accordance with MIL-STD-790.

b. The distributor shall not open the level of packaging containing the unit containers. This level of packaging shall bear a stamp stating "Not to be opened by the distributor". The distributor shall not remark or retest the product.

c. MIL-C-39006/28 product shall be handled as a nonstocking item only.

d. The distributor shall be held accountable by the manufacturer to supply the complete data package to the user with each order.

e. The manufacturer shall be responsible for auditing the distributors to the requirements of MIL-STD-790 and the limitations and controls as stated herein.

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tantalum, tubular, insulated.

Seal type: Hermetic (glass-to-tantalum anode).

Construction: Double seal.

Terminals: Axial-wire lead:

Cathode and anode (type N32, N51, or N52 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

Rated voltage: See table I.

Capacitance (cap.) nominal (nom.): See table I.

Cap. tolerance: See table I.

DC leakage (DCL) (maximum) (at 25°C): See table I.

Cap. (nom.): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (maximum): See table I.

Seal: Method 112 of MIL-STD-202, conditions C and D.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition D (500 g).

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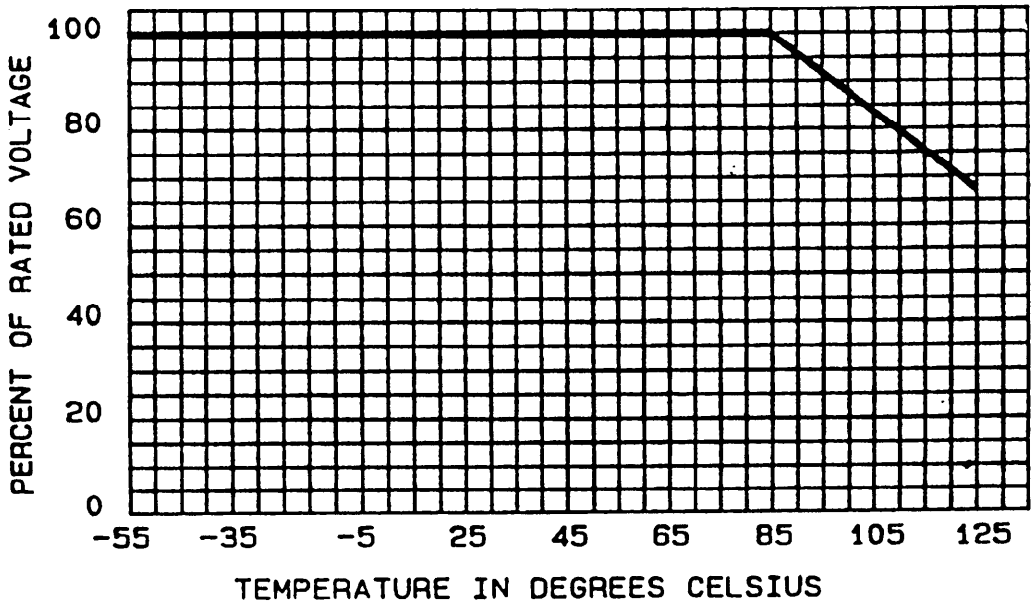


FIGURE 2. Voltage derating with temperature.

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TABLE I. Capacitor characteristics.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C and +125°C	Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
6	30	±20	1	2	9	3.98	4	6.9	100	-40	+10.5	+12	820	T1	0001
"	30	±10	"	"	"	"	"	"	100	"	"	"	820	"	0002
"	30	±5	"	"	"	"	"	"	100	"	"	"	820	"	0003
"	68	±20	"	"	15	3.16	"	"	60	"	+14	+16	960	"	0004
"	68	±10	"	"	"	"	"	"	60	"	"	"	960	"	0005
"	68	±5	"	"	"	"	"	"	60	"	"	"	960	"	0006
"	140	±20	"	3	21	1.99	"	"	40	"	"	"	1200	T2	0007
"	140	±10	"	"	"	"	"	"	40	"	"	"	1200	"	0008
"	140	±5	"	"	"	"	"	"	40	"	"	"	1200	"	0009
"	270	±20	"	6.5	45	2.21	"	"	25	-44	+17.5	+20	1375	"	0010
"	270	±10	"	"	"	"	"	"	25	"	"	"	1375	"	0011
"	270	±5	"	"	"	"	"	"	25	"	"	"	1375	"	0012
"	330	±20	2	7.9	36	1.45	"	"	20	"	+14	+16	1800	T3	0013
"	330	±10	"	"	"	"	"	"	20	"	"	"	1800	"	0014
"	330	±5	"	"	"	"	"	"	20	"	"	"	1800	"	0015
"	560	±20	"	13	55	1.30	"	"	25	-64	+17.5	+20	1900	"	0016
"	560	±10	"	"	"	"	"	"	25	"	"	"	1900	"	0017
"	560	±5	"	"	"	"	"	"	25	"	"	"	1900	"	0018
"	1200	±20	3	14	90	1.00	"	"	20	-80	+25	+25	2265	T4	0019
"	1200	±10	3	14	90	1.00	"	"	20	-80	+25	+25	2265	T4	0020
8	25	±20	1	2	7.5	3.98	5	9.2	100	-40	+10.5	+12	820	T1	0021
"	25	±10	"	"	"	"	"	"	100	"	"	+12	820	"	0022
"	25	±5	"	"	"	"	"	"	100	"	"	+12	820	"	0023
"	56	±20	"	"	14	3.32	"	"	59	"	+14	+16	900	"	0024
"	56	±10	"	"	"	"	"	"	59	"	"	+16	900	"	0025
"	56	±5	"	"	"	"	"	"	59	"	"	+16	900	"	0026
"	120	±20	"	"	20	2.21	"	"	50	-44	+17.5	+20	1220	T2	0027
"	120	±10	"	"	"	"	"	"	50	"	"	"	1220	"	0028
"	120	±5	"	"	"	"	"	"	50	"	"	"	1220	"	0029
"	220	±20	"	7	37	2.23	"	"	30	"	"	"	1370	"	0030
"	220	±10	"	"	"	"	"	"	30	"	"	"	1370	"	0031
"	220	±5	"	"	"	"	"	"	30	"	"	"	1370	"	0032

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C and +125°C	Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
8	290	±20	2	6	34	1.56	5	9.2	25	-64	+17.5	+20	1770	T3	0033
"	290	±10	"	"	"	"	"	"	"	"	"	"	1770	"	0034
"	290	±5	"	"	"	"	"	"	"	"	"	"	1770	"	0035
"	430	±20	"	14	46	1.42	"	"	"	"	"	"	1825	"	0036
"	430	±10	"	"	"	"	"	"	"	"	"	"	1825	"	0037
"	430	±5	"	"	"	"	"	"	"	"	"	"	1825	"	0038
"	850	±20	4	16	60	0.94	5	"	22	-80	+25	+25	2330	T4	0039
"	850	±10	4	16	60	0.94	5	"	22	-80	+25	+25	2330	T4	0040
10	20	±20	1	2	6	3.98	7	11.5	175	-32	+10.5	+12	820	T1	0041
"	20	±10	"	"	"	"	"	"	175	"	"	"	820	"	0042
"	20	±5	"	"	"	"	"	"	175	"	"	"	820	"	0043
"	47	±20	"	"	13	3.67	"	"	100	-36	+14	+16	855	"	0044
"	47	±10	"	"	"	"	"	"	100	"	"	"	855	"	0045
"	47	±5	"	"	"	"	"	"	100	"	"	"	855	"	0046
"	100	±20	"	4	15	1.99	"	"	60	"	"	"	1200	T2	0047
"	100	±10	"	"	"	"	"	"	60	"	"	"	1200	"	0048
"	100	±5	"	"	"	"	"	"	60	"	"	"	1200	"	0049
"	180	±20	"	7	30	2.21	"	"	40	"	"	"	1365	"	0050
"	180	±10	"	"	"	"	"	"	40	"	"	"	1365	"	0051
"	180	±5	"	"	"	"	"	"	40	"	"	"	1365	"	0052
"	250	±20	2	10	"	1.59	"	"	30	-40	"	"	1720	T3	0053
"	250	±10	"	"	"	"	"	"	30	"	"	"	1720	"	0054
"	250	±5	"	"	"	"	"	"	30	"	"	"	1720	"	0055
"	390	±20	"	16	44	1.50	"	"	25	-64	+17.5	+20	1800	"	0056
"	390	±10	"	"	"	"	"	"	25	"	"	"	1800	"	0057
"	390	±5	"	"	"	"	"	"	25	"	"	"	1800	"	0058
"	750	±20	4	"	50	0.88	"	"	23	-80	+25	+25	2360	T4	0059
"	750	±10	4	16	50	0.88	"	"	23	-80	+25	+25	2360	T4	0060

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.)		Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at			3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
			+25°C	+85°C and +125°C						-55°C	+85°C	+125°C			
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
15	15	±20	1	2	5	4.42	10	17.2	155	-24	+10.5	+12	780	T1	0061
"	15	±10	"	"	"	"	"	"	155	"	"	"	780	"	0062
"	15	±5	"	"	"	"	"	"	155	"	"	"	780	"	0063
"	33	±20	"	"	10	4.02	"	"	90	-28	+14	+16	820	"	0064
"	33	±10	"	"	"	"	"	"	90	"	"	"	820	"	0065
"	33	±5	"	"	"	"	"	"	90	"	"	"	820	"	0066
"	70	±20	"	4	13	2.46	"	"	75	"	"	"	1150	T2	0067
"	70	±10	"	"	"	"	"	"	75	"	"	"	1150	"	0068
"	70	±5	"	"	"	"	"	"	75	"	"	"	1150	"	0069
"	120	±20	"	7	18	1.99	"	"	50	"	+17.5	+20	1450	"	0070
"	120	±10	"	"	"	"	"	"	50	"	"	"	1450	"	0071
"	120	±5	"	"	"	"	"	"	50	"	"	"	1450	"	0072
"	170	±20	2	10	25	1.95	"	"	35	-32	+14	+16	1480	T3	0073
"	170	±10	"	"	"	"	"	"	35	"	"	"	1480	"	0074
"	170	±5	"	"	"	"	"	"	35	"	"	"	1480	"	0075
"	270	±20	"	16	32	1.57	"	"	30	-56	+17.5	+20	1740	"	0076
"	270	±10	"	"	"	"	"	"	30	"	"	"	1740	"	0077
"	270	±5	"	"	"	"	"	"	30	"	"	"	1740	"	0078
"	540	±20	6	24	40	0.98	"	"	23	-80	+25	+25	2330	T4	0079
"	540	±10	6	24	40	0.98	"	"	23	-80	+25	+25	2330	T4	0080
25	10	±20	1	2	4	5.31	15	28.8	220	-16	+8	+9	715	T1	0081
"	10	±10	"	"	"	"	"	"	220	"	"	"	715	"	0082
"	10	±5	"	"	"	"	"	"	220	"	"	"	715	"	0083
"	22	±20	"	"	6.6	3.98	"	"	140	-20	+10.5	+12	825	"	0084
"	22	±10	"	"	"	"	"	"	140	"	"	"	825	"	0085
"	22	±5	"	"	"	"	"	"	140	"	"	"	825	"	0086
"	50	±20	"	"	11.0	2.92	"	"	70	-28	+13	+15	1130	T2	0087
"	50	±10	"	"	"	"	"	"	70	"	"	"	1130	"	0088
"	50	±5	"	"	"	"	"	"	70	"	"	"	1130	"	0089
"	100	±20	"	10	15	1.99	"	"	50	"	"	"	1435	"	0090
"	100	±10	"	"	"	"	"	"	50	"	"	"	1435	"	0091
"	100	±5	"	"	"	"	"	"	50	"	"	"	1435	"	0092

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C and +125°C	Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
V dc	µf	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
25	120	±20	2	6	21	2.32	15	28.8	38	-32	+13	+15	1450	T3	0093
"	120	±10	"	"	"	"	"	"	38	"	"	"	1450	"	0094
"	120	±5	"	"	"	"	"	"	38	"	"	"	1450	"	0095
"	180	±20	"	18	26	1.92	"	"	32	-48	"	"	1525	"	0096
"	180	±10	"	"	"	"	"	"	32	"	"	"	1525	"	0097
"	180	±5	"	"	"	"	"	"	32	"	"	"	1525	"	0098
"	350	±20	7	28	35	1.33	"	"	24	-70	+25	+25	1970	T4	0099
"	350	±10	7	28	35	1.33	"	"	24	-70	+25	+25	1970	T4	0100
30	8	±20	1	2	4	6.64	20	34.5	275	-16	+8	+12	640	T1	0101
"	8	±10	"	"	"	"	"	"	275	"	"	"	640	"	0102
"	8	±5	"	"	"	"	"	"	275	"	"	"	640	"	0103
"	15	±20	"	"	5	4.42	"	"	175	-20	+10.5	"	780	"	0104
"	15	±10	"	"	"	"	"	"	175	"	"	"	780	"	0105
"	15	±5	"	"	"	"	"	"	175	"	"	"	780	"	0106
"	40	±20	"	5	10	3.32	"	"	65	-24	"	"	1120	T2	0107
"	40	±10	"	"	"	"	"	"	65	"	"	"	1120	"	0108
"	40	±5	"	"	"	"	"	"	65	"	"	"	1120	"	0109
"	68	±20	"	8	13	2.54	"	"	60	"	+13	+15	1285	"	0110
"	68	±10	"	"	"	"	"	"	60	"	"	"	1285	"	0111
"	68	±5	"	"	"	"	"	"	60	"	"	"	1285	"	0112
"	100	±20	2	12	17	2.26	"	"	40	-28	+10.5	+12	1450	T3	0113
"	100	±10	"	"	"	"	"	"	40	"	"	"	1450	"	0114
"	100	±5	"	"	"	"	"	"	40	"	"	"	1450	"	0115
"	150	±20	"	18	23	2.03	20	34.5	35	-48	+13	+15	1525	"	0116
"	150	±10	"	"	"	"	"	"	35	"	"	"	1525	"	0117
"	150	±5	"	"	"	"	"	"	35	"	"	"	1525	"	0118
"	300	±20	8	32	31	1.37	"	"	25	-60	+25	+25	1950	T4	0119
"	300	±10	8	32	31	1.37	"	"	25	-60	+25	+25	1950	T4	0120

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C and +125°C	Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
50	5	±20	1	2	3	7.96	30	57.5	400	-16	+5	+6	580	T1	0121
"	5	±10	"	"	"	"	"	"	400	"	"	"	580	"	0122
"	5	±5	"	"	"	"	"	"	400	"	"	"	580	"	0123
"	10	±20	"	"	4	5.31	"	"	250	-24	+8	+9	715	"	0124
"	10	±10	"	"	"	"	"	"	250	"	"	"	715	"	0125
"	10	±5	"	"	"	"	"	"	250	"	"	"	715	"	0126
"	25	±20	"	5	8	4.25	"	"	95	-20	+10.5	+12	1005	T2	0127
"	25	±10	"	"	"	"	"	"	95	"	"	"	1005	"	0128
"	25	±5	"	"	"	"	"	"	95	"	"	"	1005	"	0129
"	47	±20	"	9	11	3.11	"	"	70	-28	+13	+15	1155	"	0130
"	47	±10	"	"	"	"	"	"	70	"	"	"	1155	"	0131
"	47	±5	"	"	"	"	"	"	70	"	"	"	1155	"	0132
"	60	±20	2	12	12	2.65	"	"	45	-16	+10.5	+12	1335	T3	0133
"	60	±10	"	"	"	"	"	"	45	"	"	"	1335	"	0134
"	60	±5	"	"	"	"	"	"	45	"	"	"	1335	"	0135
"	82	±20	"	16	15	2.43	"	"	45	-32	+13	+15	1400	"	0136
"	82	±10	"	"	"	"	"	"	45	"	"	"	1400	"	0137
"	82	±5	"	"	"	"	"	"	45	"	"	"	1400	"	0138
"	160	±20	8	32	17	1.41	"	"	27	-50	+25	+25	1900	T4	0139
"	160	±10	8	32	17	1.41	"	"	27	-50	+25	+25	1900	T4	0140
60	4	±20	1	2	2.8	9.29	40	69.0	550	-16	+5	+6	525	T1	0141
"	4	±10	"	"	"	"	"	"	550	"	"	"	525	"	0142
"	4	±5	"	"	"	"	"	"	550	"	"	"	525	"	0143
"	8.2	±20	"	"	4	6.47	"	"	275	-24	+8	+9	625	"	0144
"	8.2	±10	"	"	"	"	"	"	275	"	"	"	625	"	0145
"	8.2	±5	"	"	"	"	"	"	275	"	"	"	625	"	0146
"	20	±20	"	5	7	4.64	"	"	105	-16	+10.5	+12	930	T2	0147
"	20	±10	"	"	"	"	"	"	105	"	"	"	930	"	0148
"	20	±5	"	"	"	"	"	"	105	"	"	"	930	"	0149
"	39	±20	"	9	10	3.40	"	"	90	-28	"	"	1110	"	0150
"	39	±10	"	"	"	"	"	"	90	"	"	"	1110	"	0151
"	39	±5	"	"	"	"	"	"	90	"	"	"	1110	"	0152

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C and +125°C	Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
60	50	±20	2	12	10	2.65	40	69.5	50	-16	+10.5	+12	1330	T3	0153
"	50	±10	"	"	"	"	"	"	50	"	"	"	1330	"	0154
"	50	±5	"	"	"	"	"	"	50	"	"	"	1330	"	0155
"	68	±20	"	16	13	2.54	"	"	50	-32	"	"	1365	"	0156
"	68	±10	"	"	"	"	"	"	50	"	"	"	1365	"	0157
"	68	±5	"	"	"	"	"	"	50	"	"	"	1365	"	0158
"	140	±20	8	32	16	1.52	"	"	28	-40	+20	+20	1850	T4	0159
"	140	±10	8	32	16	1.52	"	"	28	-40	+20	+20	1850	T4	0160
75	3.5	±20	1	2	2.5	9.48	50	86.2	650	-16	+5	+6	525	T1	0161
"	3.5	±10	"	"	"	"	"	"	650	"	"	"	525	"	0162
"	3.5	±5	"	"	"	"	"	"	650	"	"	"	525	"	0163
"	6.8	±20	"	"	3.5	6.83	"	"	300	-20	+8	+9	610	"	0164
"	6.8	±10	"	"	"	"	"	"	300	"	"	"	610	"	0165
"	6.8	±5	"	"	"	"	"	"	300	"	"	"	610	"	0166
"	15	±20	"	5	6	5.31	"	"	150	-16	"	"	890	T2	0167
"	15	±10	"	"	"	"	"	"	150	"	"	"	890	"	0168
"	15	±5	"	"	"	"	"	"	150	"	"	"	890	"	0169
"	33	±20	"	10	10	4.02	"	"	90	-24	+10.5	+15	1000	"	0170
"	33	±10	"	"	"	"	"	"	90	"	"	"	1000	"	0171
"	33	±5	"	"	"	"	"	"	90	"	"	"	1000	"	0172
"	40	±20	2	12	9	2.99	"	"	60	-16	"	+12	1250	T3	0173
"	40	±10	"	"	"	"	"	"	60	"	"	"	1250	"	0174
"	40	±5	"	"	"	"	"	"	60	"	"	"	1250	"	0175
"	56	±20	"	17	11	2.61	"	"	60	-28	"	+15	1335	"	0176
"	56	±10	"	"	"	"	"	"	60	"	"	"	1335	"	0177
"	56	±5	"	"	"	"	"	"	60	"	"	"	1335	"	0178
"	110	±20	9	36	12	1.45	"	"	29	-35	+20	+20	1850	T4	0179
"	110	±10	9	36	12	1.45	"	"	29	-35	+20	+20	1850	T4	0180

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C and +125°C	Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
100	2.5	±20	1	2	2	10.62	65	115	950	-16	+7	+8	505	T1	0181
"	2.5	±10	"	"	"	"	"	"	950	"	"	"	505	"	0182
"	2.5	±5	"	"	"	"	"	"	950	"	"	"	505	"	0183
"	4.7	±20	"	"	3	8.47	"	"	500	"	"	"	565	"	0184
"	4.7	±10	"	"	"	"	"	"	500	"	"	"	565	"	0185
"	4.7	±5	"	"	"	"	"	"	500	"	"	"	565	"	0186
"	11	±20	"	4	5	6.03	"	"	200	"	+8	"	835	T2	0187
"	11	±10	"	"	"	"	"	"	200	"	"	"	835	"	0188
"	11	±5	"	"	"	"	"	"	200	"	"	"	835	"	0189
"	22	±20	"	9	7.5	4.52	"	"	100	"	"	"	965	"	0190
"	22	±10	"	"	"	"	"	"	100	"	"	"	965	"	0191
"	22	±5	"	"	"	"	"	"	100	"	"	"	965	"	0192
"	30	±20	2	12	7	3.10	"	"	80	"	"	"	1240	T3	0193
"	30	±10	"	"	"	"	"	"	80	"	"	"	1240	"	0194
"	30	±5	"	"	"	"	"	"	80	"	"	"	1240	"	0195
"	43	±20	"	17	8.5	2.62	"	"	70	-20	"	"	1335	"	0196
"	43	±10	"	"	"	"	"	"	70	"	"	"	1335	"	0197
"	43	±5	"	"	"	"	"	"	70	"	"	"	1335	"	0198
"	86	±20	9	36	10	1.54	"	"	30	-25	+15	+15	1800	T4	0199
"	86	±10	9	36	10	1.54	"	"	30	-25	+15	+15	1800	T4	0200
125	1.7	±20	1	2	2	15.61	85	144	1250	-16	+7	+8	415	T1	0201
"	1.7	±10	"	"	"	"	"	"	1250	"	"	"	415	"	0202
"	1.7	±5	"	"	"	"	"	"	1250	"	"	"	415	"	0203
"	3.6	±20	"	"	2.7	9.95	"	"	600	"	"	"	520	"	0204
"	3.6	±10	"	"	"	"	"	"	600	"	"	"	520	"	0205
"	3.6	±5	"	"	"	"	"	"	600	"	"	"	520	"	0206
"	9	±20	"	5	5	7.37	"	"	240	"	"	"	755	T2	0207
"	9	±10	"	"	"	"	"	"	240	"	"	"	755	"	0208
"	9	±5	"	"	"	"	"	"	240	"	"	"	755	"	0209

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.)		Dissipation factor (max.)	ESR at 25°C (max.) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at			3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/28-
			+25°C	+85°C and +125°C						-55°C	+85°C	+125°C			
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms		
125	14	±20	1	7	6	5.69	85	144	167	-16	+7	+8	860	T2	0210
"	14	±10	"	"	"	"	"	"	167	"	"	"	860	"	0211
"	14	±5	"	"	"	"	"	"	167	"	"	"	860	"	0212
"	18	±20	2	9	5	3.69	"	"	129	"	"	"	1130	T3	0213
"	18	±10	"	"	"	"	"	"	129	"	"	"	1130	"	0214
"	18	±5	"	"	"	"	"	"	129	"	"	"	1130	"	0215
"	25	±20	"	13	6	3.18	"	"	93	"	"	"	1200	"	0216
"	25	±10	"	"	"	"	"	"	93	"	"	"	1200	"	0217
"	25	±5	"	"	"	"	"	"	93	"	"	"	1200	"	0218
"	56	±20	10	40	6.5	1.54	"	"	32	-25	+15	+15	1800	T4	0219
"	56	±10	10	40	6.5	1.54	"	"	32	-25	+15	+15	1800	T4	0220

1/ Reverse voltage rating at 85°C is 3 V dc, and at 125°C is 2 V dc.

2/ Maximum ESR is calculated by the following equation:

$$\text{ESR (max.)} = \frac{DF}{2\pi fC}, \text{ where:}$$

DF = Maximum dissipation factor from this table;
f = 120 Hz;
C = Nominal capacitance.

3/ For ripple current limits at various temperatures, voltages, and frequencies, see table II.

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TABLE II. Ripple current multipliers versus frequency, temperature, and applied peak voltage. 1/ 2/ 3/ 4/ 5/

Frequency of applied ripple current		120 Hz				800 Hz				1 kHz				10 kHz				40 kHz				100 kHz			
		Temp °C				Temp °C				Temp °C				Temp °C				Temp °C				Temp °C			
	Ambient still air	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°
6/	100%	.60	.39	---	---	.71	.43	---	---	.72	.45	---	---	.88	.55	---	---	1.0	.63	---	---	1.1	.69	---	---
	90%	.60	.46	---	---	.71	.55	---	---	.72	.55	---	---	.88	.67	---	---	1.0	.77	---	---	1.1	.85	---	---
	80%	.60	.52	.35	---	.71	.62	.42	---	.72	.62	.42	---	.88	.76	.52	---	1.0	.87	.59	---	1.1	.96	.65	---
	70%	.60	.58	.44	---	.71	.69	.52	---	.72	.70	.52	---	.88	.85	.64	---	1.0	.97	.73	---	1.1	1.07	.80	---
	≤66 2/3%	.60	.60	.46	.27	.71	.71	.55	.32	.72	.72	.55	.32	.88	.88	.68	.40	1.0	1.0	.77	.45	1.1	1.1	.85	.50

1/ At 125°C the rated voltage of the capacitors decreases to 66 2/3 percent of the 85°C rated voltage.

2/ The peak of the applied ac ripple voltage plus the applied dc voltage must not exceed the dc voltage rating of the capacitor either forward or reverse.

3/ The ripple current listed in table I represents a rating calculated by using a maximum internal temperature rise (ΔT) of 50°C at 40 kHz at 85°C ambient temperature, with a maximum peak rated voltage of 66 2/3 percent of the 85°C peak voltage rating.

4/ The maximum allowable internal temperature rise (ΔT) decreases linearly to a calculated 10°C rise at 125°C ambient temperature.

5/ The internal temperature rise is directly proportional to the ESR of the capacitor, and ESR increases with decreasing frequency.

6/ Percentage of 85°C rated peak voltage.

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Random vibration: Method 214 of MIL-STD-202, test condition II-K. The test shall be conducted for 1 hour and 30 minutes in each of three mutually perpendicular directions.

Test requirements:

a. Test potential: Rated voltage.

b. Measurements: During the last 30 minutes of vibration in each direction, an electrical measurement shall be made on the capacitors to determine intermittent open- or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption of 0.3 millisecond or greater duration.

c. Requirements during random vibration: Parts not meeting the specified limits established in MIL-C-39006/22 for post-vibration electrical measurements or which exhibit intermittent voltage spikes greater than 5 percent of the rated voltage and of 0.3 millisecond or greater in duration, or which exhibit arcing, opening, or shorting during vibration testing, shall be rejected.

d. Post-vibration requirements: DC leakage, capacitance, and dissipation factor at 25°C.

Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C).

Number of cycles: 30 for group A.

DCL: 125 percent (maximum) of 25°C value of table I for group A; 200 percent (maximum) of 25°C value of table I for group B.

ΔCap.: Within ±5 percent of initial measurement.

DF: 115 percent (maximum) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:

Pull: Method 211 of MIL-STD-202, condition A.

Wire-lead bend: In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (maximum) of +25°C value of table I.

ΔCap.: Within ±8 percent of initial measurement.

DF: 115 percent (maximum) of initial requirement.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 V dc, minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

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Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

Cap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Impedance: See table I.

ΔCap.: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap.: Within ±5 percent of step 1 value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap.: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap.: See table I.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap.: Within ±5 percent of step 1 value.

DF: See table I.

Reverse voltage: As specified in MIL-C-39006.

DCL: See table I.

Cap.: Within tolerance of table I value.

DF: See table I.

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Life at +85°C: 240 hour (group B only) - method 108 of MIL-STD-202.

DCL: See table I.

ΔCap.: Within ±10 percent of initial measurement.

DF: See table I.

AC ripple life at 85°C:

Test ripple current: See table I.

Test voltage: 66.67 percent of rated voltage (see table I).

DCL: See table I.

ΔCap.: Within ±10 percent of initial measurement.

DF: See table I.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet).

Test voltage: Rated voltage.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

DCL: See table I.

ΔCap.: Within ±5 percent of initial measurement.

DF: See table I.

Marking: In accordance with MIL-C-39006, with the following exception: The manufacturer may mark the T1 case capacitors as follows:

Example: M39006/
28-0001
JAN TM
12345
9115A

Voltage groups (85°C rated voltage):

I - - - - - - - - - -	6 through 15 V
II - - - - - - - - - -	25 through 75 V
III - - - - - - - - -	100 and 125 V

QUALIFICATION REQUIREMENTS

Qualification to this specification sheet shall consist of:

a. Current qualification to MIL-C-39006/22 at a minimum of "R" failure rate level for "H" designated units.

b. Approval of baseline document in accordance with MIL-STD-790.

c. A successful audit to the baseline.

d. Submission of groups A and B data for the first lot produced.

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QUALITY CONFORMANCE INSPECTION

Production lot. A production lot shall consist of all of the capacitors of a single nominal capacitance/voltage rating of one design. It shall be processed as a single lot through manufacturing steps on the same equipment, to the same revisions of the manufacturer's documentation, and identified with the same unique lot code. In addition, the lot shall conform to the following:

a. Tantalum powder shall be traceable to one raw material lot batch and be from the same supplier.

b. Tantalum wire shall be traceable to one raw material lot batch and be from the same supplier.

c. Production lot numbers shall provide traceability to the anode pressing batch.

d. The anode pressing batch shall be pressed within a 24 hour period on the same pressing machine. Further, it shall be sintered as a complete batch.

e. The entire production lot shall be voltage formed (same time and tank), impregnated, and otherwise processed through final sealing as a complete batch, with all parts receiving the same processing.

f. Traceability to the operator and the date each operation was performed shall be maintained.

g. Lot identity shall be maintained from the time the lot is assembled.

h. Production lot code shall be assigned at or prior to the formation of the dielectric (Ta_2O_5).

i. Traceability and test records shall be maintained for a minimum of twelve years.

Rework: Rework is not allowed on any operation except for marking, sleeving, and lead rework.

Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

Group A inspection. Group A inspection shall consist of the inspections specified in table III in the order shown. Failure to meet the applicable percent defective allowed (PDA) shall be cause for lot rejection. Summary of group A data shall be forwarded to the purchaser with parts.

TABLE III. Group A inspection.

Inspection 1/	Sampling procedure 2/
Subgroup 1	
Thermal shock	
Constant voltage conditioning 3/	
DC leakage at 25°C and 85°C 4/	100% of lot
Capacitance	
Dissipation factor	
Seal, conditions C and D	
Subgroup 2	
Stability at low and high temperatures	20 pieces 5/
Visual and mechanical inspection	
Subgroup 3	10 pieces
Solderability	0 failures

See footnotes on next page.

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TABLE III. Group A inspection - Continued.

1/ Unless otherwise specified, requirements and methods shall be in accordance with the basic specification.
2/ Overall PDA shall not exceed 5 percent.
3/ A one amp fused circuit shall be used, with a one ohm per volt maximum series resistance. Burn-in time: 168 hours minimum at +85°C.
4/ May be performed in either order, at 25°C and 85°C.
5/ Lot shall be 100 percent inspected if 1 or more failures occur for the nonconforming characteristics.

Group B inspection. Group B inspection shall consist of the tests specified in table IV and shall be performed on sample units from each production lot which has been subjected to and has passed group A inspection. Copies of group B data shall be forwarded to purchaser with parts. Production lots shall not be shipped until the conclusion of the group B inspection.

Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract.

TABLE IV. Group B inspection.

Inspection 1/	Sampling procedure
Subgroup 1 Thermal shock 2/ Surge voltage Life 3/ DC leakage (at +25°C and +85°C) Capacitance Dissipation factor Seal, conditions C and D Visual and mechanical inspection	5 samples 0 failures 4/
Subgroup 2 Thermal shock, 100 cycles Shock, specified pulse Random vibration Reverse voltage DC leakage (at +25°C and +85°C) Capacitance Dissipation factor Seal, conditions C and D Visual and mechanical inspection	5 samples 0 failures 4/

1/ Unless otherwise specified, requirements and methods are in accordance with the basic specification.
2/ 30 cycles minimum.
3/ At +85°C for 240 hours minimum. Maximum series impedance shall be 10 ohms.
4/ If a lot is rejected as a result of failures during group B inspection, the lot shall not be resubmitted.

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Environmental tests. Compliance with the following requirements is demonstrated through group C testing in MIL-C-39006/22:

Salt spray
Terminal strength
Dielectric withstanding voltage
Insulation resistance
AC ripple life
Barometric pressure
Resistance to soldering heat

PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

Baseline document. The manufacturer shall complete the baseline document and submit the required information to the qualifying activity prior to scheduling the Product and Quality Audit. It is not intended that the manufacturer submit proprietary information. Each manufacturer shall provide a detailed process flow chart which shall include, as a minimum:

a. All major process steps and inspection points by descriptive title, document number, and revision.

b. The identification of all critical manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline approval.

Changes to baseline. The audit team shall compare the latest baseline document to the previous baseline document to assure all changes have been submitted to the qualifying activity and have been approved.

a. Minor changes: Any subsequent departure from the approved flow chart (i.e., additions or deletions), or changes to the revision status of those processes or inspections are allowed, provided they do not violate the changes prohibited by section b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes: No changes (i.e., substitutions) shall be allowed in the following areas without prior review and approval of the qualifying activity. In the following areas, partial or total requalification may be required as determined by the qualifying activity.

(1) Anode design.

(2) Electrolyte formulation.

(3) Internal construction.

(4) Hermetic seal.

Audits. An initial audit and subsequent periodic audits (every year or as deemed necessary) shall be performed by the qualifying activity. The audit shall include, but not be limited to, all of the items on the checklist. The results of any additional items that are assessed shall be added to the checklist for record purposes.

Duties of the audit team.

a. The audit team shall review the baseline document for adequacy and completeness prior to the audit of the manufacturer and should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that all measuring and test equipment is functioning and calibrated. They shall also verify that test measuring procedures and equipment are sufficiently accurate for the required test.

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c. The audit team shall typically sample check 12 months of records for the item being certified. In addition, the location of the equipment and the adequacy of the operating procedure shall be checked and, whenever practical, operations utilizing the equipment and procedures witnessed.

d. The audit team shall insure that adequate documentation, trained personnel, equipment, and visual aids are available to perform the inspections specified in table V.

TABLE V. Certification checklist.

Inspection	Inspector	Satisfactory	Unsatisfactory
a. Verification of MIL-STD-790 requirements.			
b. Flow chart.			
(1) Key processes identified with revision.			
(2) Key inspection identified with revision.			
(3) Non-proprietary documents reviewed.			
(4) Screens/inspection points identified.			
c. Baseline - shall describe as a minimum the following:			
(1) Anode design.			
(2) Electrolyte formulation.			
(3) Internal construction and design.			
(4) In-process inspection.			
(5) Assembly procedures.			
(6) Sealing method.			
d. Lot control plan (see production lot definition in quality conformance inspection section of this document).			
e. Traceability plan.			
f. Training plan for QC personnel.			
g. Nondestructive test plan.			

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CONCLUDING MATERIAL

Custodians:
Air Force - 19
NASA - NA

Review activities:
Army - ER
Navy - EC
Air Force - 85
DLA - ES

Preparing activity:
Air Force - 19

Agent:
DLA - ES

(Project 5910-1728)

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